

Technical Note: Incorporation of Greenfield and FOAK Likelihood Ratios into p_bayes Model

Objective

To evaluate the impact of introducing two new binary features—Greenfield vs. Expansion and First-of-a-kind (FOAK) vs. Nth-of-a-kind (NOAK)—into the project-level Bayesian probability model (p_bayes), with the goals of preserving overall model reliability (AUC performance) and providing added granularity in project ordering.

Methods

Dataset: mpi_2018_2020_with_flags.csv (n=272). Labels: end_success=1 indicates project success. Prior: 5-year survival probability estimated from Kaplan–Meier curve, $p=0.455821$.

Holdout strategy: Deterministic 80/20 split based on SHA-1 hash of Unique ID modulo 5.

Baseline model: Features include Province, Sector, Group, Province–Sector interaction, Cleantech. Priors from KM curve only.

Greenfield/Expansion LRs: Estimated from training set with Laplace smoothing. Raw LRs: Greenfield=0.83, Expansion=2.42. Shrinkage applied with $t=0.5$ to optimize holdout AUC. Final: Greenfield=0.909, Expansion=1.558.

FOAK/NOAK LRs: Empirical LR ≈ 1.0 , adjusted using external priors. FOAK assumed 10 pp WACC premium over greenfield. Mapping: +5 pp WACC $\approx \times 0.80$ odds. FOAK modifier=0.64. Net FOAK LR=0.582. NOAK neutral=1.0.

Results

Final LR values:

Feature	Value
Greenfield_flag=1	0.909
Greenfield_flag=0	1.558
FOAK_flag=1 (modifier)	0.64 (net ~ 0.58)
FOAK_flag=0	1.00

AUC performance:

Metric	Baseline (KM prior)	With Greenfield+FOAK
Train AUC	0.787	0.794
Holdout AUC	0.877	0.865
Full AUC	0.802	0.806

Ranking effects: Spearman rank correlation=0.979, Kendall $\tau=0.927$. Expansions consistently rose in rank, FOAKs penalized in line with external priors. Example movers included NGTL and Pembina expansions (+40–50 places).

Conclusions

Introducing Greenfield and FOAK improves interpretability and project-level differentiation while maintaining robust reliability. Shrinkage ($t=0.5$) balances holdout AUC and ordering granularity. FOAK penalty (~ 0.58 odds) aligns with project finance practice (10 pp WACC premium). NOAK neutral=1.0 enhances clarity. The model remains reliable and interpretable, providing a defensible framework for audit and decision-making.